

JOSHUA CARBON

COMPUTER ENGINEER / SOFTWARE DEVELOPER

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SUMMARY

Computer Engineering graduate with hands-on experience across software development, embedded systems, IoT, AI, and automation. Built desktop applications in C# and Visual Basic using object-oriented design, alongside computer vision and embedded projects. Comfortable debugging, integrating systems end-to-end, and documenting technical work clearly.

SKILLS

Programming Languages: C#, Python, C++, C, Visual Basic

Software Development: Object-Oriented Programming (OOP), Windows Forms, GUI Development, Software Testing, Quality Assurance (QA)

Tools: Microsoft Visual Studio, Visual Studio Code, Git, Postman, Microsoft Office Suite

AI & Computer Vision: Python, OpenCV, YOLOv5, OCR, Image Processing, Roboflow, Google Colab

Embedded Systems & IoT: Arduino, Raspberry Pi, ESP32, Blynk, Sensor Integration, Embedded C

Other: MATLAB, Simulink, Proteus, Tinkercad, LTspice, Fritzing

PROJECTS

MotorPH Payroll Management System — Developed a desktop-based payroll management system using C# and Windows Forms for employee records, attendance, payroll computation, leave management, and reporting.

Inventory Management System — Developed a desktop-based inventory management application using C# and Windows Forms for maintaining product records and inventory workflows.

QA Automation Framework — Built a reusable web automation framework using Python, Robot Framework, Selenium, and SeleniumLibrary with automated test execution and reporting.

AI-Based Japanese Character Recognition using YOLOv5 — Developed a Raspberry Pi-based computer vision system using a custom-trained YOLOv5 model for real-time Japanese Kanji recognition.

Personal Portfolio Website — Built and deployed a portfolio site using React, TypeScript, and Vite, featuring React Router-based case study pages per project and a role-tailored resume selector.

EXPERIENCE

Intern/OJT — Erovoutika Robotics and Automation Solution | May 2023 – Aug 2023

- Worked alongside engineering teams to troubleshoot technical issues across automation systems, helping reduce recurring downtime and improve overall system efficiency.
- Performed debugging and calibration on automation units, ensuring consistent performance and minimizing errors during operation.
- Supported integration of sensors and embedded components into robotics systems, improving functionality and system responsiveness.
- Authored technical documentation covering setup, calibration, and troubleshooting procedures, used as a reference by the engineering team.

EDUCATION

Bachelor of Science, Computer Engineering — Mapúa University | 2019 – 2025

Bachelor of Science, Electrical Engineering | 2014 – 2019

ADDITIONAL

- Familiar with Software Development Life Cycle (SDLC) and willing to learn.
- Familiar with Object-Oriented Programming (OOP) concepts and eager to deepen understanding.
- Interested in Artificial Intelligence, Machine Learning, Automation, and related engineering technologies.